

Analysis Summary

Assumptions

- All observations and calculated episodes are independent
- Durations can be ranked and therefore I can use Mann-Whitney U and Cliffs delta
- I assumed that manual and auto opened episodes distributions have almost same shape
- Also assumed that given sample represents all IDE users

Messy data and errors handling

- I dropped all rows with missing user_id, event or timestamp
- Normalized strings to lower case, trimmed whitespaces, explicitly assigned datatype to each column
- In case there are open and close events at the same time for same user, I made close one to occur earlier to remove zero episodes
- I did not include episodes that have opening time but no closing/reopening

Strategy for matching events and distribution calculation

- I consider group table by user_id, so that we don't have any other timestamps disturbing us, also table is sorted by timestamp
- Firstly the open event will be considered, in case we get close event without open before, we dismiss it
- If after open event I get another open, I consider this as reopening, so i save episode with end-timestamp at reopening timestamp and start another episode
- If after open event I get close event, than I simply save this episode
- At the end I save user_id, open_type(manual/auto), duration_ms(episode start - episode end)

Statistical methods and findings

- I used Mann-Whitney U for measuring distributional difference. It shows that difference between manually and auto opened episode is significant

- Cliffs delta showed that random auto-opened episode is very likely to be much longer than random manually-opened one
- At Median manual episodes are 271 seconds shorter
- For median deviation and Cliffs delta I applied Bootstrapping and calculated 95% confidence interval
- IQR and Median and Standard Deviation indicated that both manually- and auto-episodes have rare very long sessions
- I also conducted an experiment in which I checked the behavior of users who both use auto and manual ways of opening. It showed that in this case auto-opened session is at median 95 seconds longer

So to sum up, automatically-opened windows tend to be opened for much longer than manually opened ones. I showed that its consistent with Cliff's delta, statistically strong with p-value and significant with median deviation. Moreover I conducted an individual based test which proved previous results.

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Mann-Whitney U: 188256.5
p: 4.299789204232955e-73
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```
Cliff's: -0.5098635736415944 95% CI: (np.float64(-0.5546511559790673), np.float64(-0.4604360306178239))
```

```
Median difference: -271.4185 95% CI: (np.float64(-324.82001249999996), np.float64(-213.59237499999995))
```

```
Summary stats (manual): {'count': 651, 'mean_s': 4640.452692780338, 'median_s': 13.998, 'std_s': 32058.624031403197, 'p25_s': 2.3935, 'p75_s': 161.994, 'max_s': 447314.
Summary stats (auto): {'count': 1180, 'mean_s': 17464.653237288137, 'median_s': 285.4165, 'std_s': 61726.33774328927, 'p25_s': 47.109, 'p75_s': 2382.6035, 'max_s': 7286
```

```
users_paired: 115
median_diff_s: -95.77650000000001
median_diff_CI: (-79902.8169, 9936.19352499957)
p_value: 3.0746212771573506e-09
```